



HOSPITALITY GROUP

Hospitality & Tourism Operations Research

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I. Executive Summary

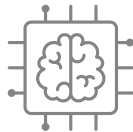
Introduction



A1 Hospitality Group is a family-owned and operated hotel property management firm based in Tri-Cities, Washington. The company has over 2 decades of experience in managing and providing services to its hotels. **A1 Hospitality Group** wants its hotels to be associated with quality, excellence, and stellar value. The group's location in the Pacific Northwest propels them to operate long-term strategically.



Current use of AI



A1 Hospitality Group currently utilizes negligible use of AI in communications with third-party owners on the corporate side. Regarding guest experience, the company has AI use in responding to reviews with AI models. This allows for **A1 Hospitality** to personalize responses back to guests, a crucial aspect of the experience.

Methods of research



Primary
Research

Site Visit:

The Courtyard Marriott Portland Airport allowed the opportunity to witness how an A1 Hotel is managed day-to-day.

Interviews:

Conducting interviews with two executives allowed for their personal perceptions of AI and findings with their jobs.

Surveys:

During Jan. 2025, 4 surveys were distributed across 3 Hotels to see the staff's perception of AI.



Secondary
Research



The secondary research comprised surfing the web to find out more about the current trends of AI in Hospitality using websites such as *Forbes* and *HotelTechReport*. Additionally, the internet also provided reasons that AI was a barrier to hotels using the technology.

Findings

Even though, AI offers many options in enhancing operations with Generative and Predictive models. The high cost has been a factor that has resulted in lower utilization.



The hospitality industry has generally faced concerns with AI compromising the human touch, an essential element that is crucial to every hotel experience



Through surveys, it was found that **A1 Hospitality** has a very moderate use of technology. Interviews also found that some staff were not as prone to technology.



Conclusions

A1 Hospitality Group can balance the costs with implementing systems that are creative or unique such as partnering with a tech partner such as, PNNL.

A1 Hospitality Group can use AI to complement the role that it plays with hotel staff in making the guest experience innovative and successful.

A1 Hospitality Group should hand out employee evaluation forms to consider the results of employees. This will foster a culture where change is welcome.

Strategic Plan



Profit:
10% Increase

Guest Satisfaction:
An average of 9/10

Employee Satisfaction:
An average of 9/10

Employee Productivity:
10% increase

G

Grasp on a inventory management system

U

Utilize energy management systems

E

Establish a chatbot for guest services

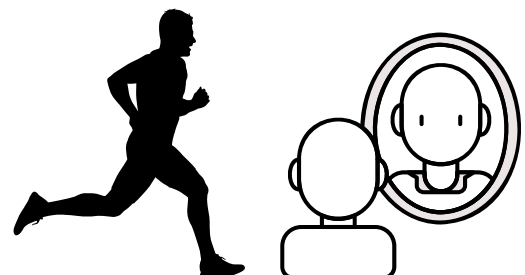
S

Speak about benefits & drawbacks with everyone

T

Transform AI to other ventures of the company

Project *GUEST* will be made in two phases: the Test Run and Evaluation. Both phases are essential to the success of project *GUEST*. The test run is crucial to testing all of the functions that the project incorporates. In the second phase, the evaluation, the focal point will be to reflect on the advantages and drawbacks. Additionally, the evaluation will work on areas to move forward in.



Quarter 1 (2025):

- Launch Inventory System

Jun-Jul '25:
Handout second survey and second meeting

Quarter 3 (2025):

- Work with PNNL to begin building an open-source model
- Launch Canary Messaging

Jan-Feb '26:
Handout final survey and final meeting

Mar-Apr '25:
Handout first survey and first meeting

Quarter 2 (2025):

- Implement thermostats throughout rooms

Sept-Oct '25:
Handout third survey and third meeting

Quarter 4 (2025):

- Begin using Clipchamp for editing videos
- Begin using open-source model

Objective	Impact	Cost
Grasp on a inventory management system	This system will cause for the inventory to be cheaper, based on the availability for items	\$40,000 (based on internet reserch)
Utilize energy management systems	One of the greater costs of a hotel is electricity, which can be targeted with using Smart Thermostats	\$50,000 (based on number of rooms multiplied by \$100 for cost of Thermostats)
Establish a chabot for guest services	A great barrier to hotels are labor costs. Chatbots will enable enhanced communication with guests. This will ultimately reduce labor costs.	\$2176 (based on \$99 per month platform fee and a additional \$2 per each room)
Transform AI in other ventures of the company	Utilizing AI in marketing to create videos and crafting a open-source tool which can identify trends	\$20,000 (based on internet research for open-source) + \$0 (Videoediting with Clipchamp is free)

Total Budget: \$112,176



II. Introduction

A. Introduction of business or organization



A1 Hospitality Group is a family-owned and operated hotel property management firm. Based in the Tri-Cities, the company proudly operates 16 premiere lodges across many states, each bearing the name of a valued entity known nationwide. This journey of advancement began humbly over 20 years ago when cofounders Mita and Vijay Patel bought a motel in eastern Oregon.



From the beginning, the objective for **A1 Hospitality Group** is for their properties to be the traveler's primary option. This target is achieved by strengthening the guest experience with consistently exceptional service provided by the employees who embrace A1 Hospitality's desire to cultivate a memorable experience from check-in to check-out. Additionally, **A1 Hospitality Group** brings travelers a great assortment of high-quality modern amenities with many of their hotels, including Free Wi-fi, fitness centers, and even culinary options provided by a restaurant or a free breakfast. All of these factors from A1 Hospitality strive to make their hotels "your home away from home"



 **Hampton Inn Seattle-Airport**
Owner at Hampton Inn Seattle-Airport

1 month ago

Hi DuckmOm,

Thank you so very much for taking the time to leave a generous review for your most recent stay with us- We're very pleased to know that you enjoyed your stay with us, and experienced the World Class Service that Hampton is known to provide. We look forward to welcoming you back in the near future.

This response is the subjective opinion of the management representative and not of Tripadvisor LLC.

The vision for a superior guest experience is further amplified by most of the ratings and reviews from customers who stay at their hotels. The figures on the left present a sample of reviews by *Tripadvisor* of the Hampton Inn Seattle Airport. Where nearly 700 reviews provide an "Excellent" category for the hotel. On the platform, most of the evaluations were followed by a reply that was personalized to fit the experience that the customer had. This process shows how A1 not only considers the feedback but also attempts to create customer loyalty with their engaging responses.



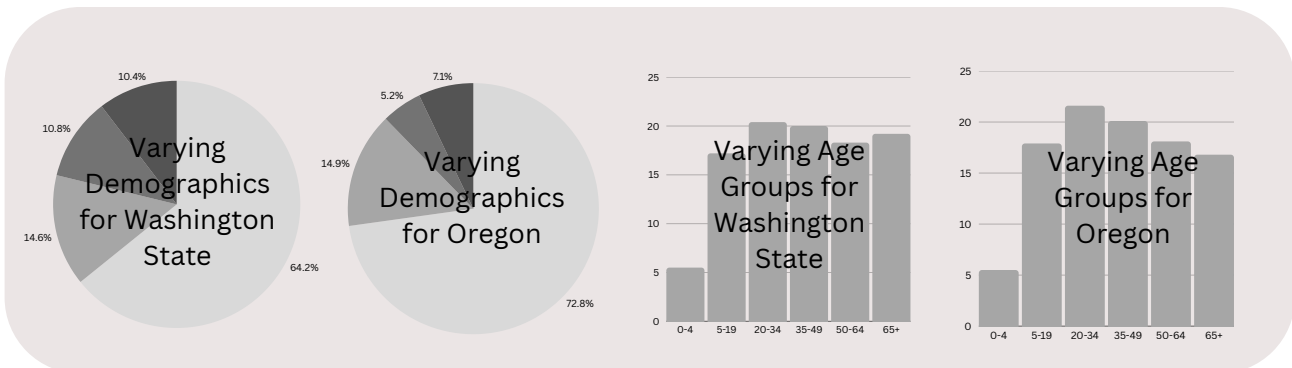
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B. Description of the Target Markets



Demographics:

A1 Hospitality has a large number of hotels, and the majority of these hotels are located in the states of Washington & Oregon. According to data from the United States Census Bureau in July of 2024, the combined population for the states of WA & OR is approximately 12.2 million. Ethnically, both states have a similar race distribution made up of White (64.2% & 72.8%), Latino or Hispanic (14.6% & 14.9%), Asian (10.8% & 5.2%), and other (10.4% & 7.1%) which includes African Americans and Native Americans, in both WA & OR respectively. Furthermore as per data collected from non-profit *USAFACTS* in 2022, the age distribution for both WA & OR are 0-4 (5.5% & 4.9%), 5-19 (17.9% & 17.2%), 20-34 (21.6% & 20.4%), 35-49 (20.1% & 20%), 50-64 (18.1% & 18.3%), 65 and older (16.8% & 19.2%), respectively. A1 Hospitality serves this diverse range of population with its properties located in areas such as Seattle, Portland, and the Tri-Cities. All these areas provide opportunities for A1 Hospitality to provide extraordinary service to all, given the area's importance to the economy.



Psychographics:

A1 Hospitality Group serves a wide base of customers psychographically. The company manages a wide variety of hotels, ranging from a Travel Lodge built for customers who are on a budget and in need of an affordable alternative to the Courtyard, which is built for business travelers who are looking for a nice bed yet with all the modern day to day needs such as free wifi or digital check-in. On the following page is an excerpt of the amenities offered at the Fairfield Hotel at Portland Airport. This lodge offers customers a comfortable set of amenities at an affordable price.

FEATURED AMENITIES ON-SITE

- | | | |
|---|--|--|
|  Sustainability |  Free Wifi |  Free Buffet and Continental and Hot Breakfast
Monday-Friday 6:00 - 9:00 AM
Saturday-Sunday 6:00 - 10:00 AM |
|  Free Coffee/Tea |  Fitness Center |  Laundry |
|  Activities for Kids |  Dry Cleaning Service |  Service Request |
|  Wake up Calls |  Digital Check In | |

This excerpt displays the varying selection of amenities at the Fairfield by Marriott by Portland International Airport

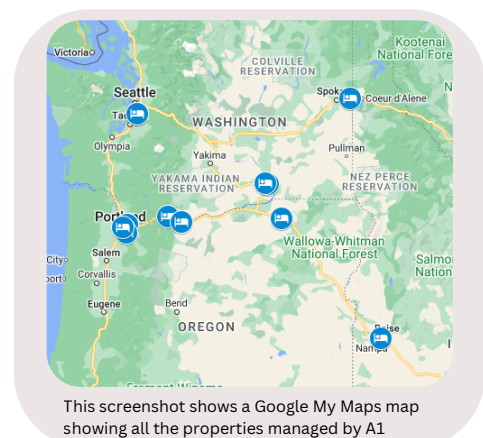
“Each of our hotels has a different target market. So some of them, for example, our hotel at the airport (Courtyard by Marriott Pasco Airport), our main target market are, the travelers who are flying in and out of the airport. Our Motel 6 that we have in Richland, our main target market are just the casual traveler looking for a quick clean bed (and) clean shower (in an) effective way. So depending on which brand it is, we tackle our marketing efforts (in a) very specific way...” -Taran Patel, Managing Principal of A1 Hospitality Group



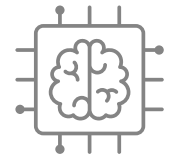
Geographics:

The bulk of operations for A1 Hospitality Group is in the innovative and rich Pacific Northwest Region, as shown right, with a screenshot from a Google My Maps page displaying the locations of all the properties managed by A1 Hospitality on the West Coast. According to *Oxford Economics*, the GDP growth of the region is expected to outpace the US by 2028. *Oxford Economics* states that economic growth has been fueled by the sectors of tech and manufacturing corporations that have grown the region to the superb jewel it is known as today. A1 Hospitality takes advantage of this growth by strategically managing its hotels near Portland, Seattle, and Boise, which are some of the largest cities in the region; this gives an opportunity to give travelers high-quality experiences with their properties located in such regions.

A1 also has many properties on Interstate 84 near the border of Washington and Oregon. These locations have been thoughtfully placed to benefit consumers who are looking for either a premium experience or a comfortable stay for a reasonable price.



C. Overview of the business or organization's current artificial intelligence strategies and practices



Guest Experience:



Regarding the guest experience, A1 Hospitality uses Artificial Intelligence to enhance the final segment of the customer's journey, as displayed in the timeline above, which is when customers create and submit reviews on various platforms such as *Google* or *Trip Advisor*. Artificial Intelligence plays a role by assisting staff in making their reviews more personal and distinctive to show the commitment to quality the organization is known for and create a personal touch to the comments or concerns the guests have moved forward.



Corporate Level:

Additionally, at the corporate level, A1 Hospitality uses Artificial Intelligence to oversee confidential conversations. The organization also uses this type of technology when submitting reports and updates at the end of the month to 3rd party ownership groups.

III. Research Methods Used in the Study

A. Description and rationale of research methodologies selected to conduct the research study



Primary Research:

Research Method:

Description:

Rationale:

Site Visit:	Visited the Courtyard by Marriott near Portland International Airport in December 2024.	This visit allowed for the opportunity to discover how A1 Hospitality manages their properties and potential areas to modify with AI.
Interviews:	Interviews Mr. Dewey Griffo, who is the Director of Sales at the Courtyard in Portland as well as Mr. Jerry Beach, VP of Operations at A1 Hospitality.	Interviews were given to both executives not only to see what it takes to run their respective business from an operational standpoint but also, their own perceptions on AI.
Survey:	We conducted a survey in January of 2025 with employees employed at across 3 hotels managed by the company	The survey was conducted to gather the views and perceptions of AI through the eyes of employees across 3 hotels. We also wanted see any suggestions they had with improving AI use.

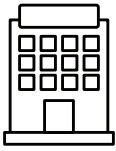


Secondary Research:

Research Method:	Description:	Rationale:
Online/Internet Research:	A variety of sites such as <i>Forbes</i> , and <i>HotelTechReport</i> were used as online research. These resources and more provided thorough insight into intergrating AI in a Hotel.	The research primarily utilized the internet and allowed us to understand different examples of how hotels are using AI to enhance their guest experience as well as the disadvantages currently facing the the industry



B. Process used to conduct selected research methods



Site Visit:

On December 31st, 2024, we had the opportunity to experience a site visit at the Courtyard Portland Airport, kindly provided by Mr. Dewey Griffo. Mr. Griffo is the sales director of the property. During our 1-hour stay, he gave us a tour of the property, giving us access to areas of the hotel, such as rooms, the *Bistro*, and the back-of-the-house operations with areas such as the kitchen and washing machines. This visit greatly shaped our view of the guest's perspective, who would be experiencing the hotel's services. On this particular day, we discovered that there were many options to increase the use of AI. One instance was with the white noise machines that were to limit noise from the outside. We saw that this particular amenity was dysfunctional. An AI system for maintenance could have fixed this particular issue.

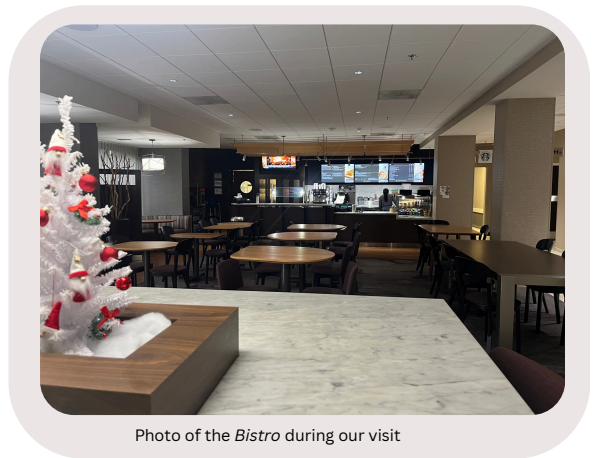


Photo of the *Bistro* during our visit

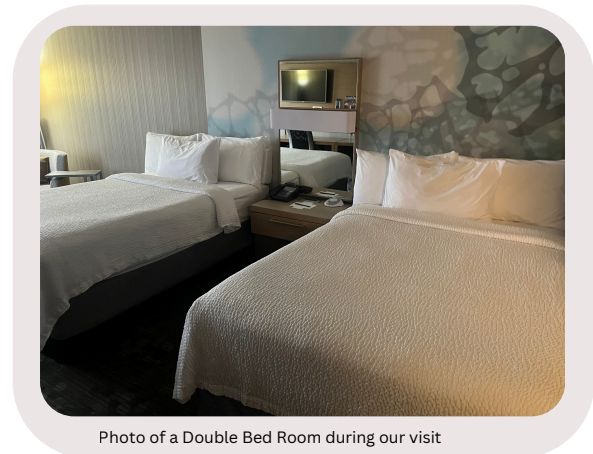


Photo of a Double Bed Room during our visit



Interviews:



On the same day as our site visit, we had the privilege to interview Mr. Dewey Griffo, Director of Sales at the Courtyard by Marriott near Portland Airport; his primary role is to employ staff and maximize the hotel's revenue. This interview gave us insight into the current basis for AI at the Courtyard and how it could grow in the future.

Later in the week, on January 2nd, Jerry Beach, Vice President of Operations for A1 Hospitality was able to connect. We chose Mr. Beach, given his extensive role in operations throughout the company and his expertise in that area. We were able to send questions and get answers from Mr. Beach. Through this interaction, we wanted to see an overview of the company and if there were any specific areas throughout the business where they could see AI intergration.



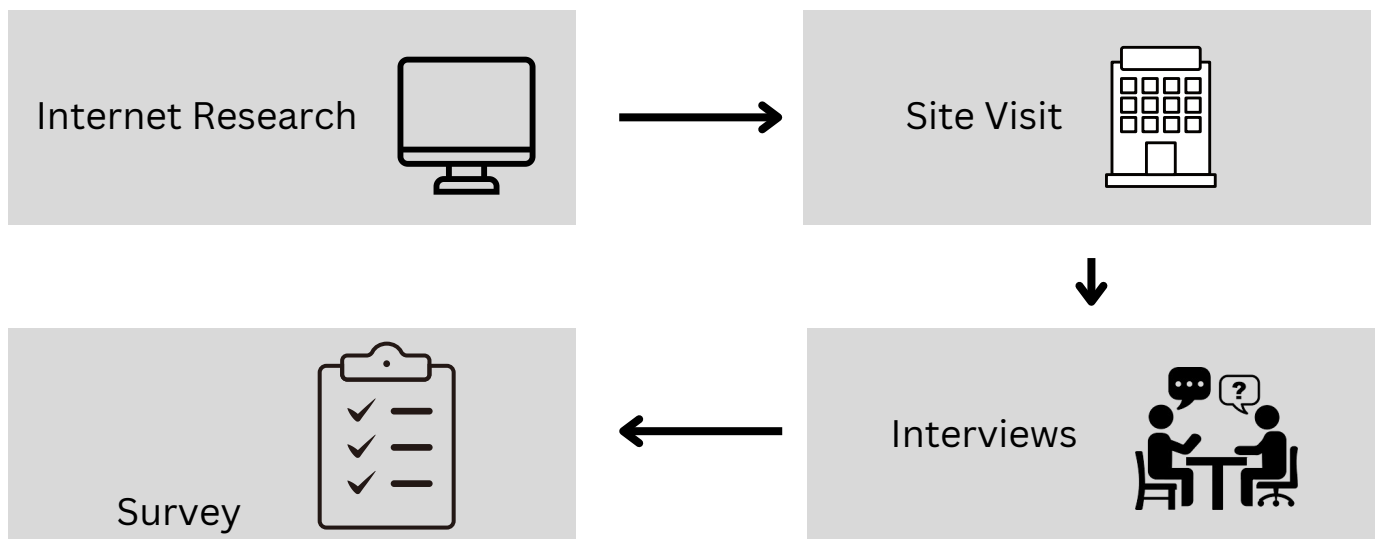
Survey:

To consider the perception of the hotel's staff, a 7-question survey was distributed through Google Forms that was answered by 4 people in varying roles. We also chose to send the survey across 3 different hotels to get a diverse amount of responses based on the level of service the hotel offered (eg. Premium or Corporate). All surveys were anonymous. The questions asked were regarding the staff's current use of technology and whether AI would assist their role.



Internet Research:

Internet Research was an essential aspect of our project. Through this method, we could catch up on trends in adding Artificial Intelligence to the guest experience in a hotel. This was very useful as it gave us background knowledge on examples of what hotels are currently optimizing in terms of AI. As well as the process and hurdles that come with these optimizations. This knowledge also provided our framework for our interviews with Mr. Griffo and Mr. Beach. Our team used a variety of websites, such as *SiteMinder*, *Forbes*, and *Revine*. Overall, the use of websites through the internet not only solidified our understanding of the use of Artificial Intelligence in hotels. It also additionally provided us with key information on how to set up our strategic plan. Additionally, information on the disadvantages of AI was also a key aspect of the research to see why hotels were not implementing the technology.



This flow chart indicates our process for our Study

IV. Findings and Conclusions of the Study

A. Findings of the Research Study

During the visit to the Courtyard Portland Airport, the primary objective was to see areas where we believed A1 Hospitality's performance could greatly improve with the advent of Artificial Intelligence. In terms of AI use, we found that A1 Hospitality had many opportunities to improve this aspect. Our team saw that in areas such as inventory, we found that these areas could have better tracking systems to minimize the waste of resources such as water or detergent. With regard to the rooms, we saw that while the room was mostly well maintained, we did see some areas that could be optimized with AI. For example, the white noise machines, which are to minimize outside noise on the bedside table, are not working. The issue itself could have been resolved if there was a system that would be able to track these fragments within the hotel and make a report of them to the necessary staff.



Photo of the laundry rooms



Photo of the bedside table, with the white noise machine



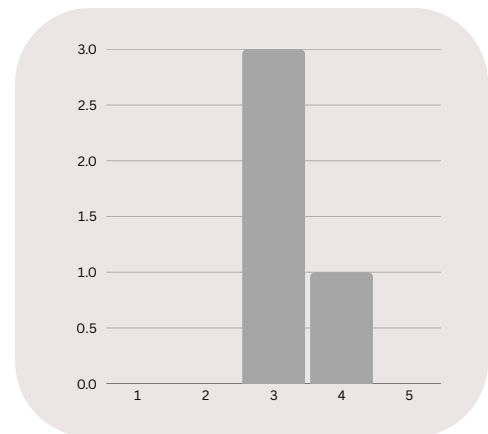
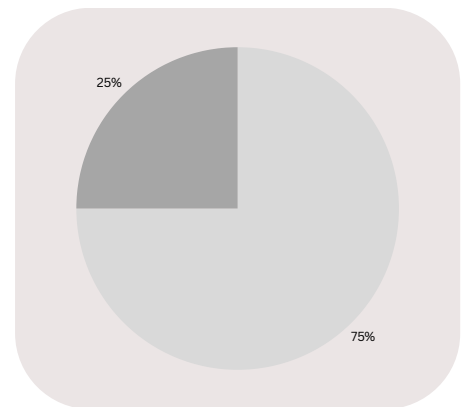
Interviews:

During the visit to the Courtyard Portland Airport, Mr. Griffo sat for an interview. During the interaction, he tended to be concerned with integrating Artificial Intelligence. This was mainly attributed to the regard of Artificial Intelligence disrupting the human touch that has been cultivated ever since hotels were established. He also alluded to self-check-in lanes that many were using. In his belief, these technologies may bring reduced hiring and less need for employees. In terms of where AI usage could be enhanced, Griffo stated that AI could help in sections such as the inventory and the use of products. However, Mr. Griffo did point to how there would be challenges in making staff transfer over to these systems, given how people are not prone to change with technology. Later in the week, when Mr. Beach was reached out too. It was discovered that A1 Hospitality has many systems managing various company operations, such as payroll and accounting. Beyond these systems, A1 Hospitality has revenue management handled by 3rd party companies. Regarding Artificial Intelligence, Mr. Beach sees much more opportunity with the technology, given its ability to cultivate much data in a matter of seconds, which he believes is beneficial to areas of the business such as marketing and revenue management. With regard to challenges facing the company. Mr. Beach stated how the organization had been impacted by the global staffing shortage as the rest of the industry has been.



Survey:

From our survey we conducted in January of 2025. We were able to have this distributed among 4 people who worked at 3 hotels. The four individuals worked in positions such as in the Restaurant, Sales, and Front Desk. This gave us a varied perspective of how they perceived AI. When staff were asked how they would rate their use of technology, out of 5. We found an average of 3.25 out of 5, which is 65%. These findings display a moderate use of technology. Additionally, we also found that in terms of technology use, the staff were using the computer to either look at transactions and emails or take room service orders. Furthermore, when asked if AI could assist in their capacity of work, where the majority said yes. When asked about areas where it could help, we found that staff presented that Artificial Intelligence could supplement their role by skillfully answering emails and having expanded knowledge of customer’s feelings.





Internet Research:

The internet provided a vast amount of information and resources on AI and its influence on the hospitality industry. Our team focused on retraining information regarding how hotels currently employ AI in their processes and what is the best way to implement it. We found throughout our research that there were two main distinctions to what AI is used, which were Generative AI and Predictive AI. Generative AI is mainly used to respond to data that has been given. While Predictive AI uses previous patterns from data points to predict future trends. In terms of ways we saw that AI was causing troubles for hoteliers was with the cost, which often is not economical for hoteliers especially medium-sized.

B. Conclusions based on findings



1. A1 Hospitality should implement a balance between cost and intergration of AI

Artificial Intelligence brings many different opportunities to A1 Hospitality. However, most solutions are expensive and not economical for hoteliers. A1 Hospitality should use creativity as well with systems that are already economical. The company can utilize chatbots or thermostats that use AI yet bring critical costs down such as labor and energy

2. A1 Hospitality must consider the Humanity Aspect

In our interview with Mr. Griffo, we found a clear emphasis on the “human touch,” which is a personalized element that helps craft a connection between employees and customers. . This detail is significant to customers maintaining loyalty towards I hotels, given that how the guest is treated will be a factor in influencing their choices. From our internet research, we found that using predictive analytics is a method where existing data can be utilized to evaluate the preferences of a customer.

3. A1 Hospitality must evaluate the feelings of staff

In a hotel, the staff are both crucial to delivering a professional service and managing the needs of the property. In the project's survey, we found that the technology usage by staff was modest but not exceptional. This is underlined by the 65% use of technology in the company by staff. Moreover, whilst interviewing Mr. Griffo, he highlighted how some of the staff were not as aware with computers as others. Providing evaluation assessments to staff will help foster a culture where staff are not only considered but also provide a voice with the progression of integrating AI.



S.W.O.T. Analysis:

Strengths:

- Established Company with over 2 decades of experience
- Emphasize Importance of guest experience and quality
- Awards from WSU Carson College of Business and ORLA
- Based in one of the fastest growing regions in the country

Weaknesses:

- Limited use of AI with chatbots for responses
- Multiple systems that make it difficult to streamline operations
- Loss of Human Touch due to inability to pick up nuances of humans
- Amount of time to set up and train all the staff

Opportunities:

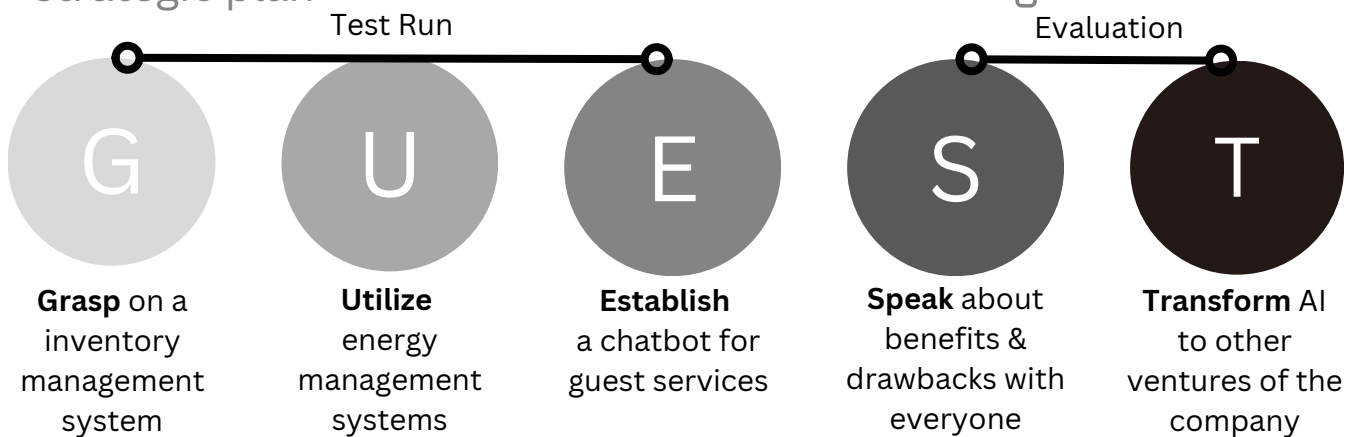
- Enhancing the guest experience with predictive analytics and chatbots
- Automating the management of supplies to ensure the optimal amount used
- Innovating maintenance systems to ensure better tracking of amenities
- Utilizing dynamic pricing to guarantee greater revenue management

Threats:

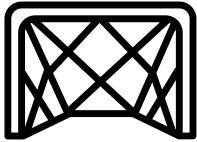
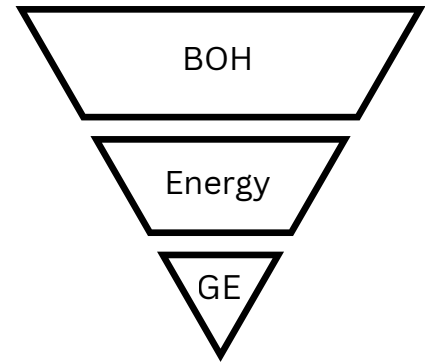
- Relying on 3rd party companies for communication and data handling
- Tracking data that can compromise privacy and data theft
- High startup costs and training with employees
- Inexperienced staff with technology

V. Proposed Strategic Plan

A. Objectives and rationale of the proposed strategic plan



To bring forward the utilization of Artificial Intelligence for A1 Hospitality, the project *GUEST* was formed. This fresh approach will commence in 5 stages, which will be divided into 2 phases: Test Run & Evaluation. Through these methods, Project *GUEST* will be able to provide innovation targeted at Guests. The structure of the plan is around 3 key functions crucial to hotel operations that are arranged in a pyramid, as seen on the right.



Objectives:

Phase 1: Test-Run

Objective #1: Grasp on an inventory management system

Rationale: Establish a system that will effectively ensure optimal supplies are used

- A1 Hospitality does not utilize any system that measures the proper inventory needed
- There is also a lack of predictive systems that are used

Solution:

- Using a predictive model with retracted data will determine optimal trends for inventory management
- This will reduce costs associated with importing supplies or food
- Maintenance will be tracked much better

Objective #2: Utilize energy management systems

Rationale: Saving crucial electrical costs to create more sustainable practices

- Energy Costs are some of the highest in the hotel industry
- These costs are formed by water usage or electricity

Solution:

- Installing smart thermostats will reduce these costs
- Creates a sustainable advantage, which is what eco-travellers are looking for

Objective #3: Establish a chatbot for guest services

Rationale: Use predictive analytics and chatbots to enhance the Guest Experience

- AI is commonly seen as a barrier to the “Human Touch,” an essential trait of Hospitality
- This is because of the models not being able to understand the nuances of human behavior

Solution:

- AI can be used to improve the experience of the guests through the use of Chatbots
- Utilizing Chatbot systems such as Canary Technologies Messaging will be able to enhance the Guest Experience by using generative AI to answer questions that guests have
- Labor costs will also be cut

Phase 2: Reflection**Objective #4: Speak about benefits & drawbacks with everyone**

Rationale: Establish a culture where change is welcome and reflect on benefits and drawbacks with everyone

- A1 Hospitality's staff currently have moderate technology use
- Some staff within the company aren't as prone to technology use
- A1 Hospitality lacks communication with staff and management

Solution:

- A survey will be sent to employees to complete
- This will allow A1 Hospitality to analyze the advantages and disadvantages of AI from the perception of their staff
- A management meeting to discuss what should stay with the plan and what should be adapted

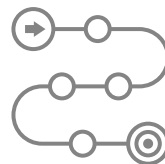
Objective #5: Transform AI to other ventures of the company

Rationale: Expanding AI in other methods, with Open-Source AI or Marketing

- AI presents a multitude of opportunities to the Hospitality Industry
- A1 Hospitality currently faces an issue with the employee crisis that has been left by the COVID-19 pandemic
- AI presents a multitude of opportunities to the Hospitality Industry
- A1 Hospitality currently faces an issue with the employee crisis that has been left by the COVID-19 pandemic

Solution:

- A1 Hospitality can take advantage of video editors using AI to market across social media to hire employees
- A1 Hospitality can partner with PNNL to develop open-source solutions

B. Proposed activities and timelines

To cultivate a practical plan that works on innovating the space of hotels with the magic of AI, Project *GUEST+* will be rolled out in two phases: a Test run and an

Evaluation series. Both phases will evolve in multiple steps. This will allow AI to be targeted to specific aspects that are essential to the operations of hotels to ultimately streamline these areas and make enhancements to the guest experience. The phases and steps are outlined below:

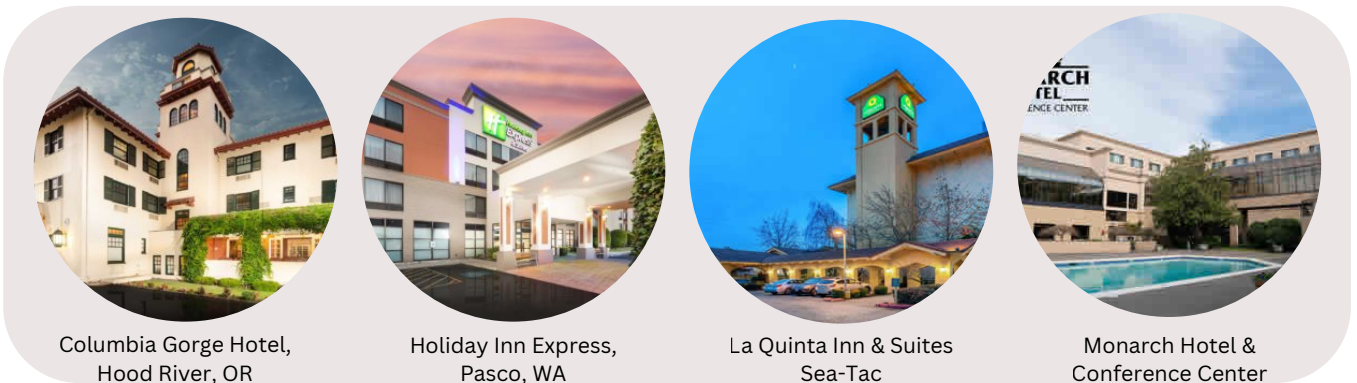
Phase 1: Test-Run

- Highlighted in first 4 steps
- The first phase involves experimenting with AI in a selection of Hotels
- This will determine what areas work well and not as much

Phase 2: Evaluate

- Used in last 2 steps
- The second phase involves bringing everyone (Employees and Management) to the table to evaluate the results of the First Phase
- A1 Hospitality will then work with the results found to advance AI in other areas

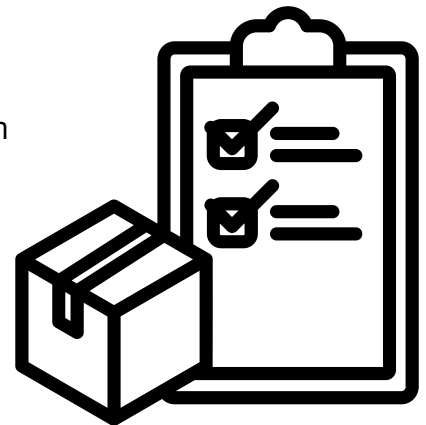
Selection of Hotels:



Project *GUEST* will be based in four locations with varying target markets (Budget to Premium). This will allow A1 Hospitality to analyze the varying characteristics of Artificial Intelligence in their hotels. The Marriott Courtyard Portland Airport was excluded due to an ongoing deal with Delta Inn Hotel Group.

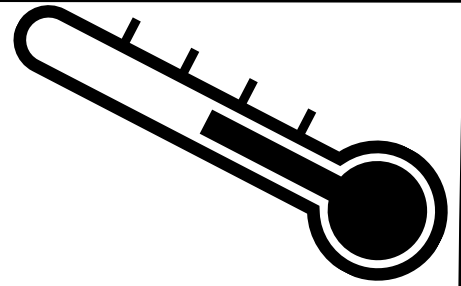
Grasp on an inventory management system

Rolling in Project *GUEST* will include introducing an inventory management system. Including this type of system will enable A1 Hospitality to leverage AI into tracking the optimal amount of supplies that are needed based on a predictive model, allowing for more precision. Taking a grasp of inventory management will allow hotels being able to reduce their costs for supplies and have additional supplies in case of maintenance issues.



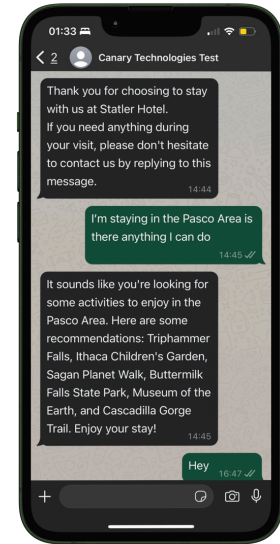
Utilize energy management systems

Sustainability is a metric all hotels strive for. It will not only bring in eco-friendly travelers but also, allow for less costs with energy. Utilizing energy management systems through smart thermostats will allow AI to control the heating in a room based on if someone is in the room.



Establish a chatbot for guest services

Chatbots will be used to meet the customers' needs. Canary Messaging chatbot utilizes generative AI for upsells and personalized recommendations through WhatsApp and Messages. Canary promises increases of upsell revenue by 250%. The company also has stated reduced reliability on staff. An example with WhatsApp is on the right.



Speak about benefits & drawbacks with everyone

Kicking off phase two will involve creating a survey that will be shared with staff across the hotel periodically. Later, a meeting with the staff in A1 Hospitality will be held to discuss the project's results based on the metrics on page 18.

Additionally, an analysis of the surveys will also be discussed. Together, this step will provide a beacon for A1 Hospitality to cultivate a more welcoming culture of change.



Transform AI to other ventures of the company

The final segment of Project GUEST will be to enhance AI through other methods in marketing and open-source AI. Using Microsoft Clipchamp's video editor, A1 Hospitality can create detailed videos through generative AI, which can be used to attract potential employees through various social media. Partnering with the Pacific Northwest National Lab (PNNL) based in Tri-Cities, who already use Artificial Intelligence for the environment. A1 Hospitality can work to build a system that builds on guest preferences and is tailored to them.

Timeline:

Quarter 1 (2025):

- Launch Inventory System

Mar-Apr '25:
Handout first survey and first meeting

Jun-Jul '25:
Handout second survey and second meeting

Quarter 2 (2025):
• Implement thermostats throughout rooms

Quarter 3 (2025):

- Work with PNNL to begin building an open-source model
- Launch Canary Messaging

Sept-Oct '25:
Handout third survey and third meeting

Jan-Feb '26:
Handout final survey and final meeting

Quarter 4 (2025):

- Begin using Clipchamp for editing videos
- Begin using open-source model

C. Proposed metrics or key performance indicators to measure plan effectiveness



Metrics	HOTR_WA_Ha que_	Performance	Method of Improvement
Profit	Increase profit based on performance of systems	+10% profit in one year	Having less costs with labor and energy. Additionally, ancillary revenue is increased with Chatbots
Guest	Increase the satisfaction of guests	9/10 on average given to employees	Having surveys distributed to Guests
Employee	Increase the employee confidence and revenue per employee	9/10 on surveys given by employees, increase revenue per employee by 10%	Having survey distributed and using PMS

VI. Proposed Budget

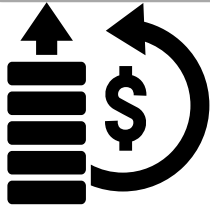
Objective	Impact	Cost
Grasp on a inventory management system	This system will cause for the inventory to be cheaper, based on the availability for items	\$40,000 (based on internet reserch)
Utilize energy management systems	One of the greater costs of a hotel is electricity, which can be targeted with using Smart Thermostats	\$50,000 (based on number of rooms multiplied by \$100 for cost of Thermostats)
Establish a chabot for guest services	A great barrier to hotels are labor costs. Chatbots will enable enhanced communication with guests. This will ultimately reduce labor costs.	\$2176 (based on \$99 per month platform fee and a additional \$2 per each room)

Transform AI in other ventures of the company

Utilizing AI in marketing to create videos and crafting a open-source tool which can identify trends

\$20,000 (based on internet research for open-source) +
\$0 (Videoediting with Clipchamp is free)

Total Budget: \$112,176



Return on Investment:

Given that financial results from **A1 Hospitality Group** were not permitted, there is not a return on investment, however with reduced costs in labor, energy, and inventory as well as ancillary upsells with numbers ranging from 10-20% for reduction in sales. The estimated ROI is around 20%-30%.

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